الجامعة المصرية
للتعلم الإلكتروني الأهلية

EELU
THE EGYPTIAN E-LEARNING UNIVERSITY

Advanced Computer Networks Section 1

Introduction to Wireshark

What is Wireshark?

A tool that captures packets passing through the network interface.

Function:

Displays the contents of protocol fields in captured messages

Main Components:

- **Packet Capture Library (pcap)**: Receives a copy of every link-layer frame
- **Packet Analyzer**: Displays details of all protocol fields

2. How to Use Wireshark

1.Open Wireshark and select Network Interface (Wi-Fi or Ethernet)
2. Start Packet Capture
3. Generate Network Traffic (visit a specific website)
4. Stop Capture
5. Use Display Filters (e.g., http)

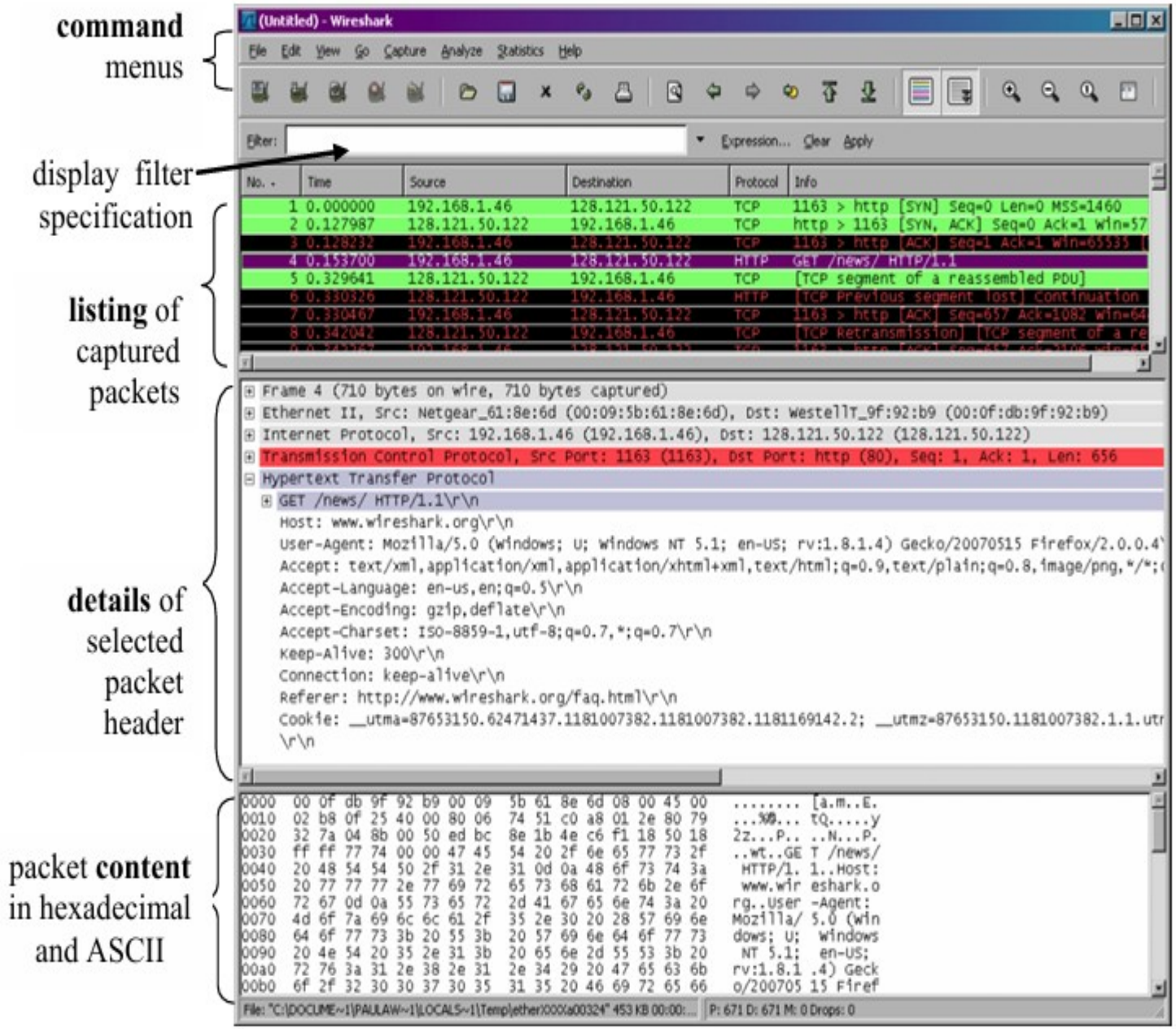


Figure 3: Wireshark Graphical User Interface, during packet capture and analysis

Layer 4 - Transport Layer (TCP/UDP)

Important Fields:

- Source Port Number
- Destination Port Number
- Sequence Number
- Acknowledgment Number
- TCP Flags (SYN, ACK, FIN)
- Checksum

Layer 3 - Network Layer (IP)

Important Fields:

- Source IP Address (Logical Address)
- Destination IP Address (Logical Address)
- Protocol Type (TCP=6, UDP=17)
- TTL (Time to Live)
- Header Length
- Total Length

Layer 2 - Data Link Layer (Ethernet)

Important Fields:

- Source MAC Address (Physical Address)
- Destination MAC Address (Physical Address)
- Ethernet Type (IPv4 = 0x0800)
- Frame Check Sequence (FCS)

Questions

1. From the Wireshark trace, identify which protocol is operating at each layer?

2. What are the Port Numbers used in the Transport Layer? Are they Well-Known Ports?

3. What protocol is operating at Layer 3 (Network Layer)?

4. From the Wireshark trace, identify the following addresses:

1. Logical Address (IP) (Source,Destination)
2. Physical Address (MAC) (Source,Destination)

5. What standard is being used at the Data Link Layer?